

hardware development is an iterative process, using industry-ready boards at this stage will definitely save time and money. With a huge variety of boards to select from, engineers can do their due diligence and select one whose microcontroller chip can be used in their PCBs after successful testing. Thus, industry-ready boards can be used as a temporary solution to reduce time to market.

For your minimum viable product (MVP). Although the full form says that it is a product, more often than not companies treat their MVP as a part of their development process. Most hardware startups do not prefer to invest too much money in the first version of anything that launches into the market. In such a scenario, using an industry-ready board will not only benefit the company financially but also ensure a quality product. Similar to the case of alpha and beta testing, engineers can migrate to custom PCBs once they are sure of the product's maturity on the field. In this case, these boards provide an interim solution that allows the R&D team to understand the viability of the technology they have developed.

When to use in end products

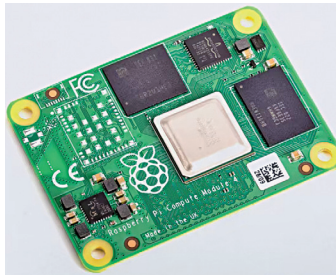
By definition, development boards contain peripherals and circuitry that allow engineers to experiment and prototype. Generally, development boards are not used in appliances because they contain circuitry that may not be required in the final product. This affects the battery life and efficiency of the device. Hence, a custom PCB is made at this stage.

That being said, almost all industry-ready boards are tailored for a particular application. If selected wisely according to the requirements of the final product, they can even be used as a permanent solution. Using them is easier if the final product is to be stationary and connected to a continuous power supply, like a gateway node in Industrial IoT (IIoT) applications. Nevertheless, rigorous

quality assurance must be done based on the environment in which the final product is to be launched.

Ten industry-ready boards

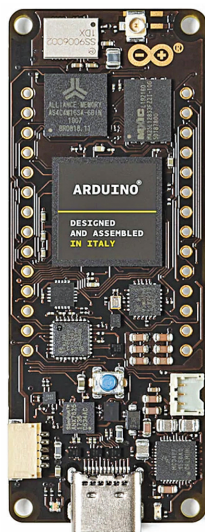
1. Raspberry Pi Compute Module 4 (CM4002008). This module is particularly designed for end applications. It has a powerful compute engine that is far superior to



(Source: Raspberry Pi)

its predecessors. It is also the first to offer an external antenna connector, which is vital in industrial applications. This is because the circuitry is always present in an enclosure, which inhibits Wi-Fi connectivity. But here, we can directly connect a certified antenna (also sold by Raspberry Pi) and enjoy a really good Wi-Fi connection.

2. Portenta H7 (ABX00042). It is Arduino's industry-grade development board that intends to satisfy the needs and demands of Industry 4.0. It has a wide range of pins

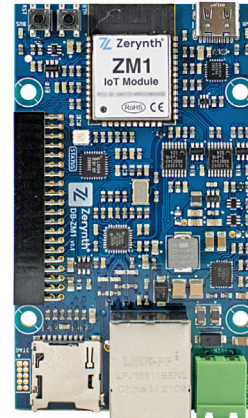


(Source: Arduino)

and can be programmed using high-level languages including C++ (Arduino framework), JavaScript, and MicroPython. An interesting feature is that it can run TensorFlow for machine learning. Considering all its characteristics, the Portenta H7

is well suited for robotics applications.

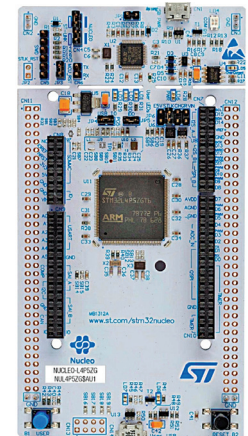
3. ZM1-DB. This board is a complete package for IoT applications. It contains an ESP based microcontroller, which makes it easy for developers with prior experience



(Source: Zerynth)

in ESP8266 or ESP32 to start using this board. With its industry-grade hardware and IoT cloud platform (Zerynth Cloud), ZM1-DB can be viewed as a useful solution for prototyping as well as direct deployment. It contains an interface standard called zBus, which allows us to easily expand the board's capabilities using various expansion boards provided by Zerynth. Therefore, it is heavily customisable and can be tailored to any end application.

4. NUCLEO-L4P5ZG. This board from STMicroelectronics provides a cost-effective and energy-efficient solution to developers. The STM32 ecosystem and communities provide a good platform for understanding this board and learning how to use it. It integrates an on-board



(Source: STMicroelectronics)

debugger/programmer and can be programmed using various IDEs. It is suitable for applications with many peripherals since all STM32 input-output pins are available for use. It